

Extending the Oracle Approximating Shrinkage
to General Distribution

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Abstract

The Oracle Approximating Shrinkage (OAS) was created assuming an underlying Gaussian distribution. In this work, we extend the tool to all distributions with 4th moments.

1 The formulation

This work should be seen as an extension of [1]. To set the stage, we first port over the relevant equations.

$$\rho_O = \frac{E \left[\text{Tr} \left(\left(\Sigma - \hat{S} \right) \left(\hat{F} - \hat{S} \right) \right) \right]}{E \left[\|\hat{S} - \hat{F}\|_F^2 \right]} \quad (1)$$

$$E \left[\text{Tr} \left(\left(\Sigma - \hat{S} \right) \left(\hat{F} - \hat{S} \right) \right) \right] = \frac{\text{Tr}(\Sigma)}{p} E \left[\text{Tr}(\hat{S}) \right] - \frac{E \left[\text{Tr}^2(\hat{S}) \right]}{p} - E \left[\text{Tr}(\Sigma \hat{S}) \right] + E \left[\text{Tr}(\hat{S}^2) \right] \quad (2)$$

$$E \left[\|\hat{S} - \hat{F}\|_F^2 \right] = E \left[\text{Tr}(\hat{S}^2) \right] - \frac{E \left[\text{Tr}^2(\hat{S}) \right]}{p} \quad (3)$$

$$E \left[\text{Tr}(\hat{S}) \right] = \text{Tr}(\Sigma) \quad (4)$$

It is at this point, we will deviate slightly from [1] by considering the general distribution case. Observe that

$$\begin{aligned} E \left[\text{Tr}(\hat{S}^2) \right] &= E \left[\|\hat{S}\|_F^2 \right] \\ &= \sum_{i,j} \frac{1}{n^2} \sum_{k,l} E \left[X_i^k X_j^k X_i^l X_j^l \right] \\ &= \sum_{i,j} \frac{1}{n^2} \left(\sum_{k \neq l} E \left[X_i^k X_j^k X_i^l X_j^l \right] + \sum_{k=l} E \left[X_i^k X_j^k X_i^l X_j^l \right] \right) \\ &= \frac{n-1}{n} \sum_{i,j} \Sigma_{i,j}^2 + \frac{1}{n^2} \sum_{i,j} \sum_k E \left[(X_i^k)^2 (X_j^k)^2 \right] \\ &= \frac{n-1}{n} \text{Tr}(\Sigma^2) + \frac{1}{n} \sum_{i,j} E \left[X_i^2 X_j^2 \right] \end{aligned}$$

Similarly,

$$\begin{aligned}
E \left[\text{Tr}^2(\hat{S}) \right] &= E \left[\sum_{i,j} S_{ii} S_{jj} \right] \\
&= \sum_{i,j} \frac{1}{n^2} \sum_{k,l} E \left[(X_i^k)^2 (X_j^l)^2 \right] \\
&= \sum_{i,j} \frac{1}{n^2} \left(\sum_{k \neq l} E \left[(X_i^k)^2 (X_j^l)^2 \right] + \sum_{k=l} E \left[(X_i^k)^2 (X_j^k)^2 \right] \right) \\
&= \frac{n-1}{n} \sum_{i,j} \Sigma_{ii} \Sigma_{jj} + \frac{1}{n^2} \sum_{i,j} \sum_k E \left[(X_i^k)^2 (X_j^k)^2 \right] \\
&= \frac{n-1}{n} \text{Tr}^2(\Sigma) + \frac{1}{n} \sum_{i,j} E \left[X_i^2 X_j^2 \right]
\end{aligned}$$

In both cases, the term $E \left[X_i^2 X_j^2 \right]$ shows up. For a centered general distribution with a 4th moment, we have

$$E \left[X_i^2 X_j^2 \right] = \Sigma_{ii} \Sigma_{jj} + 2 \Sigma_{ij}^2 + \kappa(X_i, X_i, X_j, X_j) \quad (5)$$

where $\kappa(X_i, X_i, X_j, X_j)$ represents the fourth-order joint cumulant. Note that this cumulant vanishes for a gaussian distribution. Thus,

$$\sum_{i,j} E \left[X_i^2 X_j^2 \right] = \text{Tr}^2(\Sigma) + 2 \text{Tr}(\Sigma^2) + \sum_{i,j} \kappa(X_i, X_i, X_j, X_j) \quad (6)$$

Let $\alpha = \sum_{i,j} \kappa(X_i, X_i, X_j, X_j)$.

Returning to the above,

$$E \left[\text{Tr}(\hat{S}^2) \right] = \frac{n+1}{n} \text{Tr}(\Sigma^2) + \frac{1}{n} \text{Tr}^2(\Sigma) + \frac{\alpha}{n} \quad (7)$$

$$E \left[\text{Tr}^2(\hat{S}) \right] = \text{Tr}^2(\Sigma) + \frac{2}{n} \text{Tr}(\Sigma^2) + \frac{\alpha}{n} \quad (8)$$

which is very similar to the results in [1] with the extra $\frac{\alpha}{n}$ term. Collating the

results above gives us

$$\rho_O = \frac{(1 - 2/p) \text{Tr}(\Sigma^2) + \text{Tr}^2(\Sigma) + \alpha(1 - 1/p)}{(n + 1 - 2/p) \text{Tr}(\Sigma^2) + (1 - n/p) \text{Tr}^2(\Sigma) + \alpha(1 - 1/p)} \quad (9)$$

2 Extending OAS

With the relevant formulation out of the way, we now consider the original design of OAS, but extended to include our new term $\alpha(1 - 1/p)$. As in [1], we have

$$\hat{\rho}_{j+1} = \frac{(1 - 2/p) \text{Tr}(\hat{\Sigma}_j \hat{S}) + \text{Tr}^2(\hat{\Sigma}_j) + \alpha(1 - 1/p)}{(n + 1 - 2/p) \text{Tr}(\hat{\Sigma}_j \hat{S}) + (1 - n/p) \text{Tr}^2(\hat{\Sigma}_j) + \alpha(1 - 1/p)} \quad (10)$$

$$\hat{\Sigma}_{j+1} = (1 - \hat{\rho}_{j+1})\hat{S} + \hat{\rho}_{j+1}\hat{F} \quad (11)$$

which leverages (9), but replaces $\text{Tr}(\Sigma)$ and $\text{Tr}(\Sigma^2)$ with $\text{Tr}(\hat{\Sigma}_j)$ and $\text{Tr}(\hat{\Sigma}_j \hat{S})$, respectively. Let

$$\hat{\phi} = \frac{\text{Tr}(\hat{S}^2) - \text{Tr}^2(\hat{S})/p}{(1 - 2/p) \text{Tr}(\hat{S}^2) + \text{Tr}^2(\hat{S})} = \frac{\text{Tr}(\hat{S}^2) - \text{Tr}^2(\hat{S})/p}{M} \quad (12)$$

and $\beta = \alpha(1 - 1/p)/M$. Plugging in Σ_j from (11) and plugging into (10) yields

$$\hat{\rho}_{j+1} = \frac{1 + \beta - (1 - 2/p)\hat{\phi}\hat{\rho}_j}{1 + \beta + n\hat{\phi} - (n + 1 - 2/p)\hat{\phi}\hat{\rho}_j} \quad (13)$$

Using the change of variables

$$b_j = \frac{1}{1 + \beta - (n + 1 - 2/p)\hat{\phi}\hat{\rho}_j} \Leftrightarrow \hat{\rho}_j = \frac{1 + \beta - \frac{1}{b_j}}{(n + 1 - 2/p)\hat{\phi}}, \quad (14)$$

we obtain (10) is equivalent to

$$b_{j+1} = \frac{n\hat{\phi}}{1 + \beta - (1 - 2/p)\hat{\phi}} b_j + \frac{1}{1 + \beta - (1 - 2/p)\hat{\phi}} \quad (15)$$

It is easy to see that

$$\lim_{j \rightarrow \infty} b_j = \begin{cases} \infty & \frac{n\hat{\phi}}{1 + \beta - (1 - 2/p)\hat{\phi}} \geq 1 \\ \frac{1}{1 + \beta - (n + 1 - 2/p)\hat{\phi}} & \frac{n\hat{\phi}}{1 + \beta - (1 - 2/p)\hat{\phi}} < 1 \end{cases} \quad (16)$$

This implies

$$\lim_{j \rightarrow \infty} \hat{\rho}_j = \begin{cases} \frac{1 + \beta}{(n + 1 - 2/p)\hat{\phi}} & (n + 1 - 2/p)\hat{\phi} \geq 1 + \beta \\ 1 & (n + 1 - 2/p)\hat{\phi} < 1 + \beta \end{cases} \quad (17)$$

or equivalently,

$$\rho_{OAS_{extended}} = \min \left(1, \frac{1 + \beta}{(n + 1 - 2/p)\hat{\phi}} \right) \quad (18)$$

Note that, in the limit, we have $1 + \beta \geq 0$ (The proof is deferred to Appendix A).

What's left is a consistent estimator for the β term. Given that $M = (1 - 2/p) \text{Tr}(\hat{S}^2) + \text{Tr}^2(\hat{S})$ is already handed, this leaves the α term. Given its definition and (6), the natural choice for the estimator would be

$$\hat{\alpha} = \left(\sum_{i,j} \frac{1}{n} \sum_k (X_i^k)^2 (X_j^k)^2 \right) - \text{Tr}^2(\hat{S}) - 2 \text{Tr}(\hat{S}^2), \quad (19)$$

which converges to the true α by the law of large numbers. However, for finite

samples, $\hat{\alpha}$ may drift from α , and thus recommend clamping the result, i.e.

$$\hat{\rho}_{OASextended} = \max \left(0, \min \left(1, \frac{1 + \beta}{(n + 1 - 2/p)\hat{\phi}} \right) \right) \quad (20)$$

A Proof That $1 + \beta \geq 0$

It is obvious from (18) that $\rho_{OAS_{extended}} \leq 1$, but leaves open the question of the lower bound. For this we analyze the numerator $1 + \beta$, and find that it is non-negative.

Proof. We observe $\sum_{i,j} E[X_i^2 X_j^2] = E[\|X\|^4]$ and $\text{Tr}^2(\Sigma) = (E[\|X\|^2])^2$. Thus,

$$\alpha = \sum_{i,j} E[X_i^2 X_j^2] - \text{Tr}^2(\Sigma) - 2 \text{Tr}(\Sigma^2) = \text{Var}(\|X\|^2) - 2 \text{Tr}(\Sigma^2)$$

implying $\alpha \geq -2 \text{Tr}(\Sigma^2)$. Noting $\beta = \alpha(1 - 1/p)/M$, it suffices to show

$$-2 \text{Tr}(\Sigma^2) \cdot \frac{1 - 1/p}{M} \geq -1 \Rightarrow M \geq 2(1 - 1/p) \text{Tr}(\Sigma^2).$$

Substituting $M = (1 - 2/p) \text{Tr}(\Sigma^2) + \text{Tr}^2(\Sigma)$, this reduces to

$$\text{Tr}^2(\Sigma) \geq \text{Tr}(\Sigma^2).$$

For any positive semidefinite Σ with eigenvalues $\lambda_1, \dots, \lambda_p \geq 0$,

$$\text{Tr}^2(\Sigma) - \text{Tr}(\Sigma^2) = \left(\sum_i \lambda_i \right)^2 - \sum_i \lambda_i^2 = 2 \sum_{i < j} \lambda_i \lambda_j \geq 0.$$

Hence $1 + \beta \geq 0$, and $\rho_{OAS_{extended}} \in [0, 1]$ for the true α . $\square$

References

- [1] Y. Chen, A. Wiesel, Y. C. Eldar, and A. O. Hero, "Shrinkage algorithms for MMSE covariance estimation," *IEEE Transactions on Signal Processing*, vol. 58, no. 10, pp. 5016–5029, Oct. 2010.